Measurement as Dynamical Sector Separation in Closure Dynamics

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Abstract

Paper XXX proposed that outcome probabilities arise as geometric weights over observer-admissible sectors, with the Born rule recovered from the stationary sector measure of the closure dynamics under three assumptions: equilibrium, well-separated sectors, and sector-supported decomposition. The present paper addresses the most critical of these: *why are outcome sectors well-separated?*

This paper proposes that measurement interactions dynamically reshape the closure landscape, producing multiple dynamically separated, apparatus-correlated basins corresponding to distinct outcomes. The mechanism is: system–apparatus coupling introduces an interaction term $\mathcal{F}_{\text{int}}$ into the total closure functional, deforming an effectively unseparated landscape into one with multiple dynamically separated minima and high inter-basin barriers. When the barrier height satisfies $\Delta\mathcal{F}_{ij} \gg \sigma^2$, Kramers-type transition rates between basins are exponentially suppressed, overlap and cross terms between sector-supported wavefunctions become negligible, and outcomes are dynamically stabilised.

A toy model demonstrates the mechanism: a neutral apparatus with a single quadratic minimum, when coupled to a two-state system, yields two apparatus-correlated separated basins whose barrier height grows with the coupling strength. A timescale argument shows that sector formation can occur faster than measurement readout under generic conditions.

The paper does not claim a complete solution to the measurement problem. It identifies a dynamical route by which the sector-separation assumption of Paper XXX arises from the structure of measurement interactions, without requiring external state reduction, and without formulating the mechanism in terms of environmental tracing.

1 Introduction

Paper XXX [Smart, 2026c] showed that, in the equilibrium closure regime, the Born rule can be recovered from the stationary sector measure of the closure dynamics. The argument depends on three assumptions:

1. The system is at or near the Paper XIV equilibrium.
2. The outcome sectors are well-separated basins with negligible overlap.
3. The Nelson wavefunction admits a decomposition into sector-supported components.

Assumption 1 is justified by the ergodic convergence of the stochastic closure dynamics (Paper XIV [Smart, 2026a]). Assumption 3 is natural once assumption 2 holds and sector-localised modes can be defined. The critical assumption is therefore 2: *why should outcome sectors be well-separated?*

The gap

In standard quantum mechanics, the transition from superposition to definite outcomes is typically attributed to decoherence: interaction with the environment suppresses off-diagonal elements of the density matrix. The VFD framework does not use density matrices or environmental tracing. The question is whether the closure dynamics provides an intrinsic mechanism for sector separation.

Proposal

This paper proposes that measurement interactions dynamically deform the closure functional, producing well-separated basins. The mechanism is geometric: the interaction term reshapes the landscape into multiple basin regions separated by high barriers. Sector separation is not assumed; it arises from the structure of the measurement coupling.

Scope

The paper develops the formal framework, works a toy model, provides a timescale argument, and connects the result to Paper XXX. It does not claim a complete solution to the measurement problem, does not derive decoherence from the closure framework, and does not specify the interaction functional from first principles. It identifies a dynamical route to sector separation and shows that this route justifies the assumptions of Paper XXX under stated conditions.

2 Programme Position

The logical chain:

1. **Structure**: closure geometry on the 600-cell (Papers I–V, XXII).
2. **Dynamics**: stochastic closure process with stationary distribution (Paper XIV).
3. **Quantum recovery**: Schrödinger equation as equilibrium tangent limit (Papers XVIII–XXI).
4. **Observer**: constraint substructure determining admissible configurations (Paper XXIX).
5. **Measurement**: dynamical sector separation via landscape deformation (*this paper*).

6. **Probability**: stationary sector measure yields Born weighting (Paper XXX).

Steps 5 and 6 were reversed in the order of publication: Paper XXX appeared first because the probability construction is logically downstream, but it depends on the sector-separation result developed here. This paper fills the gap between steps 4 and 6.

3 Formal Setup

3.1 Configuration space

Let $\Phi_S \in \mathcal{H}_S$ denote the system configuration and $\Phi_A \in \mathcal{H}_A$ the apparatus configuration, where $\mathcal{H}_S$ and $\mathcal{H}_A$ are the respective configuration spaces. The total configuration is $\Phi = (\Phi_S, \Phi_A) \in \mathcal{H}_S \times \mathcal{H}_A$.

3.2 Closure functional

Definition 1 (Total closure functional)**.** *The total closure functional for a measurement context is:*

$$\mathcal{F}_{\text{total}}[\Phi_S, \Phi_A] = \mathcal{F}_S[\Phi_S] + \mathcal{F}_A[\Phi_A] + \mathcal{F}_{\text{int}}[\Phi_S, \Phi_A] \tag{1}$$

where $\mathcal{F}_S$ is the system closure functional, $\mathcal{F}_A$ is the apparatus closure functional, and $\mathcal{F}_{\text{int}}$ is the interaction term encoding the measurement coupling.

Remark 2. *Before measurement, $\mathcal{F}_{\text{int}} = 0$ and the system and apparatus evolve independently. Measurement begins when the interaction is switched on ($\mathcal{F}_{\text{int}} \neq 0$). The interaction term does not destroy the closure structure; it deforms the landscape.*

3.3 Stochastic dynamics

The total configuration evolves under the stochastic closure dynamics (Paper XIV):

$$d\Phi = -\nabla \mathcal{F}_{\text{total}}\, dt + \sigma\, dW \tag{2}$$

where $\sigma > 0$ is the noise strength and dW is a Wiener process. The stationary distribution is:

$$P_{\text{st}}(\Phi) \propto e^{-2\mathcal{F}_{\text{total}}(\Phi)/\sigma^2} \tag{3}$$

4 Pre-Measurement Landscape

Before the interaction is switched on ($\mathcal{F}_{\text{int}} = 0$), the total functional is separable:

$$\mathcal{F}_{\text{total}} = \mathcal{F}_S[\Phi_S] + \mathcal{F}_A[\Phi_A] \tag{4}$$

If the system is in a superposition-like state, $\mathcal{F}_S$ has a single broad minimum (or multiple shallow, poorly separated minima). The apparatus $\mathcal{F}_A$ has its own equilibrium configuration (the "ready state"). The total landscape has a single effective basin.

Pre-measurement: single basin, no sector separation, interference present.

5 Measurement Interaction and Landscape Deformation

When the measurement interaction is switched on, $\mathcal{F}_{\text{int}}$ couples the system and apparatus configurations. The interaction term introduces correlations: the closure cost of the total configuration depends on the joint state of system and apparatus.

5.1 Correlation structure

A measurement interaction is characterised by the property that it energetically favours configurations in which the apparatus state is correlated with the system state. Formally:

Definition 3 (Measurement-type interaction)**.** *An interaction $\mathcal{F}_{\text{int}}[\Phi_S, \Phi_A]$ is of measurement type if:*

1. *It has N low-cost channels: for each system eigenstate $\Phi_S^{(i)}$, there exists a corresponding apparatus state $\Phi_A^{(i)}$ such that $\mathcal{F}_{\text{int}}[\Phi_S^{(i)}, \Phi_A^{(i)}]$ is small.*

2. *Mismatched configurations have high interaction cost: $\mathcal{F}_{\text{int}}[\Phi_S^{(i)}, \Phi_A^{(j)}] \gg \mathcal{F}_{\text{int}}[\Phi_S^{(i)}, \Phi_A^{(i)}]$ for $i \neq j$.*

Remark 4. *This is the closure-dynamical analogue of the standard requirement that a measurement apparatus correlates its pointer states with the system's eigenstates. The VFD version is expressed in terms of the closure landscape rather than Hilbert-space operators.*

5.2 Landscape deformation

The effect of a measurement-type interaction on the total closure landscape is:

Proposition 5 (Sector-forming measurement interactions)**.** *Suppose $\mathcal{F}_{\text{int}}$ is of measurement type (Definition 3), the correlated channels $(\Phi_S^{(i)}, \Phi_A^{(i)})$ are nondegenerate local minima of $\mathcal{F}_{\text{total}}$ under standard regularity assumptions, and the mismatch costs are sufficiently large. Then the total functional $\mathcal{F}_{\text{total}}$ exhibits N dynamically separated basin regions centred at $\Phi^{(i)*} = (\Phi_S^{(i)}, \Phi_A^{(i)})$, with inter-basin barriers satisfying:*

$$\Delta\mathcal{F}_{ij} = \mathcal{F}_{\text{total}}(\textit{saddle between } i, j) - \mathcal{F}_{\text{total}}(\Phi^{(i)*}) \geq \mathcal{F}_{\text{int}}[\Phi_S^{(i)}, \Phi_A^{(j)}] \tag{5}$$

Proof (schematic). Each correlated pair $(\Phi_S^{(i)}, \Phi_A^{(i)})$ is a local minimum of $\mathcal{F}_{\text{total}}$ because the system and apparatus terms are minimised separately and the interaction cost is small (by the low-cost channel condition). The nondegeneracy assumption ensures these are genuine local minima with positive-definite Hessian. Any path from minimum i to minimum j must pass through a configuration where either the system or apparatus is mismatched, incurring interaction cost $\mathcal{F}_{\text{int}}[\Phi_S^{(i)}, \Phi_A^{(j)}]$ or $\mathcal{F}_{\text{int}}[\Phi_S^{(j)}, \Phi_A^{(i)}]$. The saddle-point energy is bounded below by this mismatch cost. A rigorous proof would require specifying the topology of the configuration space and verifying transversality conditions on the critical points. □

> Measurement interaction: effectively unseparated landscape → N separated basins. The barrier height is set by the mismatch cost of the interaction.

6 Transition Suppression

6.1 Kramers-type barrier crossing

Paper XIV [Smart, 2026a] established that inter-basin transition rates in the stochastic closure dynamics follow Kramers-type estimates:

$$\Gamma_{i\to j} \sim e^{-2\Delta\mathcal{F}_{ij}/\sigma^2} \tag{6}$$

When $\Delta\mathcal{F}_{ij} \gg \sigma^2$, the transition rate is exponentially suppressed.

Definition 6 (Sector separation condition)**.** *The outcome sectors are well-separated if:*

$$\Delta\mathcal{F}_{ij} \gg \sigma^2 \qquad \textit{for all } i \neq j \tag{7}$$

Remark 7. *This is not introduced here as an independent postulate; under the regularity, nondegeneracy, and large-mismatch conditions of Proposition 5, it follows from the measurement-type interaction structure.*

6.2 Cross-term suppression

When sectors are well-separated, the Nelson wavefunction $\Psi = \sum_i c_i\phi_i$ (where ϕ_i is supported near minimum i) has the property that the sector-supported components have effectively disjoint supports. The cross terms satisfy:

$$\int_{\Phi_i} \phi_k^* \, \phi_l \, d\mu \approx 0 \qquad \text{for } k \neq l \tag{8}$$

because the overlap between ϕ_k (localised near minimum k) and ϕ_l (localised near minimum l) is exponentially small in the barrier height:

$$|\langle\phi_k|\phi_l\rangle| \lesssim e^{-\Delta\mathcal{F}_{kl}/\sigma^2} \tag{9}$$

Interference between sectors is suppressed by the same mechanism that suppresses transitions: the barrier height.

7 Outcome Stability

Once the system has settled into basin i, it remains there with high probability. The stability has two sources:

1. **Barrier suppression.** The Kramers rate $\Gamma_{i\to j} \sim e^{-2\Delta\mathcal{F}_{ij}/\sigma^2}$ is exponentially small, so spontaneous transitions out of the basin are extremely rare.

2. **Observer constraint reinforcement.** The observer $(\mathcal{O}, \mathcal{C}_\mathcal{O})$ of Paper XXIX [Smart, 2026b] provides an additional stability mechanism: the observer's coherence constraint $\mathcal{C}_\mathcal{O} \leq \epsilon$ may be satisfied in basin i but not in the transition region between basins. The observer constraint acts as an additional barrier.

   Classical definiteness of measurement outcomes corresponds to stable basin occupancy in the deformed closure landscape.

8 Timescale Argument

For the sector-separation mechanism to produce definite outcomes, the basins must form before the measurement result is read.

Definition 8 (Separation timescale)**.** *The separation timescale τ_{sep} is the time required for the closure landscape to deform from a single-basin to a multi-basin configuration after the interaction $\mathcal{F}_{\text{int}}$ is switched on. As a curvature-based dimensional proxy, one may estimate:*

$$\tau_{\text{sep}} \sim \frac{1}{\|\nabla^2 \mathcal{F}_{\text{int}}\|} \tag{10}$$

the inverse of the curvature scale set by the interaction.

Remark 9 (Timescale hierarchy)**.** *For the mechanism to work, the following hierarchy must hold:*

$$\tau_{\text{sep}} \ll \tau_{\text{relax}} \ll \tau_{\text{obs}} \tag{11}$$

where τ_{relax} is the time for the stochastic dynamics to settle into a basin (set by the local relaxation rate of $\mathcal{F}_{\text{total}}$) and τ_{obs} is the timescale on which the observer reads the outcome. The first inequality says basins form before the system relaxes into one; the second says the system has settled before the result is recorded. This hierarchy is treated as a working hypothesis for macroscopic measurement apparatus; whether it holds universally is an open question.

9 Heuristic Status of the Basin-Formation Argument

The present paper identifies a *mechanism class* for sector formation, not a universal theorem for all measurements. The argument establishes that measurement-type interactions (Definition 3), under regularity and nondegeneracy conditions, produce separated basins with high barriers. This is a sufficient condition, not a necessary one: alternative routes to sector separation may exist. The interaction functional $\mathcal{F}_{\text{int}}$ is introduced phenomenologically; its derivation from first principles or from the 600-cell geometry is deferred. The toy model below is illustrative, not a claim that all measurement interactions take this specific form.

10 Toy Model: Coupled Double-Well Formation

This model is illustrative. It is not derived from the full 600-cell construction, but demonstrates the landscape-deformation mechanism in the simplest possible setting.

Example 10 (System–apparatus coupling)**.** ***System.*** *One-dimensional, with closure functional $\mathcal{F}_S(\phi) = \alpha(\phi^2 - 1)^2$, having minima at $\phi = \pm 1$.*

Apparatus. *One-dimensional, with closure functional $\mathcal{F}_A(\chi) = \frac{1}{2}\kappa\chi^2$, having a single minimum at $\chi = 0$ (the "ready state").*

Interaction. *$\mathcal{F}_{\text{int}}(\phi, \chi) = \frac{1}{2}\lambda(\chi - g\phi)^2$, favouring apparatus state $\chi = g\phi$ (the apparatus tracks the system).*

Total functional.

$$\mathcal{F}_{\text{total}}(\phi, \chi) = \alpha(\phi^2 - 1)^2 + \tfrac{1}{2}\kappa\chi^2 + \tfrac{1}{2}\lambda(\chi - g\phi)^2 \tag{12}$$

Pre-measurement *($\lambda = 0$). The system has two minima at $\phi = \pm 1$; the apparatus sits at $\chi = 0$. The total landscape has two minima at $(\pm 1, 0)$, but the apparatus does not distinguish them.*

Post-measurement *($\lambda > 0$). Minimising over χ at fixed ϕ: $\chi^* = \lambda g \phi/(\kappa + \lambda)$. The effective system functional becomes:*

$$\mathcal{F}_{\text{eff}}(\phi) = \alpha(\phi^2 - 1)^2 + \frac{\kappa \lambda g^2}{2(\kappa + \lambda)} \phi^2 \tag{13}$$

The two minima shift and the total landscape has two well-separated minima at:

$$(\phi, \chi)^{(1)*} \approx (+1,\ +g\lambda/(\kappa + \lambda)), \qquad (\phi, \chi)^{(2)*} \approx (-1,\ -g\lambda/(\kappa + \lambda)) \tag{14}$$

Barrier. *The saddle between the two minima occurs near $\phi = 0$, $\chi = 0$, with barrier height:*

$$\Delta \mathcal{F} \approx \alpha + \frac{\kappa \lambda g^2}{2(\kappa + \lambda)} \cdot 1 \tag{15}$$

which grows with the coupling strength λ and the tracking gain g.

Sector separation. *For $\Delta \mathcal{F} \gg \sigma^2$, the transition rate between basins is $\Gamma \sim e^{-2\Delta \mathcal{F}/\sigma^2} \approx 0$. The apparatus state χ now records the system outcome: $\chi > 0$ means "system in $\phi = +1$ basin"; $\chi < 0$ means "system in $\phi = -1$ basin."*

Conclusion. *In this toy model, the system already possesses two minima, but the measurement interaction promotes them from unregistered system alternatives to dynamically separated, apparatus-correlated outcome sectors. The coupling ($\lambda > 0$) deforms the closure landscape from a state where the apparatus is neutral into a state where the apparatus is correlated with the system and the two outcomes are dynamically separated by a high barrier.*

11 Connection to Paper XXX

Paper XXX [Smart, 2026c] derived Born-like weighting $P_i = |c_i|^2$ under three assumptions. The present paper addresses the most critical:

1. **Well-separated sectors** (Assumption 2 of Paper XXX). This paper shows that measurement-type interactions (Definition 3) produce barrier heights satisfying $\Delta \mathcal{F}_{ij} \gg \sigma^2$ (Proposition 5), which is the sector-separation condition (Definition 6).

2. **Sector-supported decomposition** (Assumption 3 of Paper XXX). When sectors are well-separated, the wavefunctions ϕ_i localised in each basin have effectively disjoint supports (equation 9), so the decomposition $\Psi = \sum_i c_i \phi_i$ is valid with negligible cross terms.

3. **Equilibrium** (Assumption 1 of Paper XXX). This paper does not address the equilibrium assumption. It remains as stated in Paper XXX: a working hypothesis justified by the ergodic convergence of the closure dynamics.

   The well-separated sector assumption used in Paper XXX arises dynamically from measurement-induced deformation of the closure landscape.

Table 1: Structural comparison: closure sector separation vs. standard decoherence.

Feature	Standard decoherence	VFD sector separation
Mechanism	Environmental entanglement	Landscape barrier formation
Interference loss	Off-diagonal density matrix decay	Cross-term suppression by barrier
Pointer states	Einselection (environment-preferred)	Basin minima of $\mathcal{F}_{\text{total}}$
Outcome stability	Decoherence timescale	Kramers barrier-crossing rate
Environment	Required (traced out)	Not required in this mechanism class
Formalism	Density matrices, partial trace	Closure functional, gradient flow

12 Comparison to Standard Decoherence

Remark 11 (Structural analogy, not equivalence)**.** *The two frameworks are structurally analogous: both produce effective sector separation and interference suppression. The key structural difference is that standard decoherence requires tracing over environmental degrees of freedom, while closure sector separation arises from the geometry of the total closure landscape without invoking an external environment. Whether the two mechanisms are related by a deeper correspondence (e.g. whether environmental coupling can be recast as landscape deformation) is an open question.*

13 Discussion and Limitations

13.1 What is established

- Measurement-type interactions are defined in terms of closure-landscape structure (Definition 3).
- Under regularity and nondegeneracy conditions, such interactions produce dynamically separated basins with high barriers (Proposition 5).
- Barrier height suppresses inter-basin transitions exponentially (Kramers estimate, equation 6).
- Cross terms between sector-supported wavefunctions are suppressed by the same barrier mechanism (equation 9).
- Outcome stability has two sources: barrier suppression and observer-constraint reinforcement.
- A toy model demonstrates the mechanism explicitly (Example 10).
- A timescale hierarchy is identified as a working hypothesis (equation 11).
- The well-separated sector assumption of Paper XXX is justified under the stated conditions.

13.2 What is not established

- No proof that all physically relevant measurement interactions are of measurement type (Definition 3).

- No derivation of $\mathcal{F}_{\text{int}}$ from first principles or from the 600-cell geometry.
- No proof of the timescale hierarchy (11) for general measurement contexts.
- No treatment of continuous-spectrum measurements or measurements with degenerate outcomes.
- No full comparison to standard decoherence theory (the comparison in Table 1 is structural, not quantitative).
- No proof that the barrier mechanism is the *only* route to sector separation; alternative mechanisms may exist.
- The interaction functional $\mathcal{F}_{\text{int}}$ is introduced phenomenologically, not derived from the constraint structure.

14 Conclusion

Measurement, in this framework, is not external state reduction. It is the dynamical formation of separated outcome sectors through landscape deformation. When a system couples to a measurement apparatus, the interaction term reshapes the closure functional into a multi-basin landscape of apparatus-correlated outcome sectors. The barrier height between basins is set by the mismatch cost of the interaction; when this height exceeds σ^2, transitions are exponentially suppressed, cross terms are suppressed to negligible size, and outcomes are dynamically stabilised.

This result provides the missing structural justification for Paper XXX. The well-separated sector assumption is not an unexplained postulate; it is a consequence of measurement-type interactions deforming the closure landscape. Together with Paper XXX's stationary-measure argument, this completes a two-step route from measurement interaction to Born-like probability:

1. **This paper**: measurement interaction $\to$ sector separation $\to$ stable basins.
2. **Paper XXX**: stable basins $\to$ stationary sector measure $\to$ Born weighting.

The chain is conditional: it depends on the measurement interaction being of the right type, the barrier height being sufficient, the timescale hierarchy holding, and the system being near equilibrium. These conditions are stated explicitly. A complete, assumption-free measurement theory remains an open target. What is established is a dynamical route from interaction to outcome, without invoking state reduction, environmental tracing, or an independent probabilistic measurement postulate.

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